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# Facile ultrasonic-assisted one-step preparation of sugarcane bagasse carbon sorbent for bio-based odor removal cat litter formulation

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ABSTRACT

To reduce the cost of adsorbents for the removal of cat urine odor there has been concerted research efforts to produce functionalized porous carbon from waste biomass. In this work, a cost-effective and environmentally friendly method was used to prepare a carbon adsorbent (AC-SNP) from sugarcane bagasse through ultrasonic-assisted impregnation of mixed $HNO_3/H_3PO_4$ on bagasse prior to carbonization. Ultrasonication enhanced structural deformation of bagasse and enriched the N and O-containing groups prior to cabonization which allowed the final carbon material to have increased specific surface area of 1603.1 $m^2/g$ and porosity of 1.3858 $cm^3 \cdot g^{-1}$ (including micro- and meso-pores), and increased surface functionality of -COOH, -OH, pyrrolic N, and pyridinic N groups. The adsorption capacity of the carbon material for 3-mercapto-3-methyl-1-butanol (MMB), a volatile cat urine component, was up to 207.12 mg/g at 25 °C, and the heterogenous multi-layer Freundlich model and Pseudo-second-order model were used to explain the adsorption mechanism. Density functional theory (DFT) calculations showed that there are electrostatic interactions and hydrogen bonding between the O- and N-containing groups and MMB molecules. AC-SNP was found to be nontoxic in a cat litter formulation, and removes 99.87% volatile MMB at a low loading ratio of 2 wt%. The fabricated sugarcane bagasse-based cat litter (SB_CL) formulation also has anti-bacterial properties, and so has significant commercialization prospects.

## 1. Introduction

Cats have been the main companion animal in many countries because of the ease of care, small enough to be kept easily in small living spaces such as apartments and are also able to endure long separations from their owners without apparent problems (Bernstein, 2007). In 2020, the number of cat owners in China was 27.01 million, with a year-on-year increase of 13.33%. Owners love their cats, however, cat urine odor is a problem for many people as the odor is unpleasant. Volatile organosulfur compounds (VOSCs) including 3-mercapto-3-methyl-1-butanol (MMB), 3-mercapto-3-methylbutyl-formate, 3-methyl-3-methylthio-1-butanol, and 3-methyl-3-(2-methyldisul-fanyl)-1-butanol have been identified in cat urine as responsible for the unpleasant odor (Miyazaki et al., 2006), and have the potential to induce streaks and negative emotions in humans when over=inhaled (Shugart, 2005). Therefore, it is of great importance to develop odor removal cat litter formulations to minimize the unpleasant odors.

Rremoval of VOSCs with adsorbents is considered a good approach because of the advantages of simple operation and high efficiency (Guo et al., 2021). Carbon materials have been widely used as adsorbents due

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to their excellent adsorption capacities stemming from large specific surface area, adjustable porosity and abundant functional groups (Ding et al., 2018). Low cost biomass wastes such as chitin, oil palm ash, fruit hulls, corncob, and bagasse etc. are used as the feedstocks for the production of carbons used for the removal of different environmental contaminants (Marrakchi et al., 2017; Khanday et al., 2016; Islam et al., 2017; Luana et al., 2017; Jawad et al., 2021). Physical activation with air, $CO_2$ or steam, and chemical activation through impregnating biomass precursor with $H_3PO_4$, NaOH/KOH, and $ZnCl_2$ prior to carbonization are the two approaches used to improve the textural, and surface functional properties of carbon (Gupta et al., 2013; Islam et al., 2017b,a; Khanday et al., 2019; Saka et al., 2012). In a recent study carried out by Vaughn et al. (2020), a bio-based odor-reducing, low dust-forming, clumping cat litter was prepared from Eastern red cedar (ERC) wood fibers and ERC-derived biochar (pyrolyzed at 928 °C for 24 min), with which the headspace levels of MMB (a cat urine model compound) was reduced by 59.3% with the addition of 10% biochar to the litter formulation. The relative low removal efficiency of MMB by the biochar was because it did not have the desired properties for MMB adsorption. In a previous study by the same authors (Vaughn et al., 2011), copper sulfate which has been demonstrated to be a cat urine odor reducing agent was added to a formulation (0.25 mg/g) to assist in improving performance. Nonetheless only a small increment of 16% was achieved. Furthermore, constant exposure to Cu (a heavy metal) may cause health-related issues to cats as the LD50 for copper sulfate orally ingested in mammals is reported as 470 mg/kg (Gosselin et al., 1984). Therefore, it is necessary to fabricate an environmentally friendly material such a carbon derived from biomass with the desired properties for the removal of felinine malodor compounds present in cat litter formulations. To endow the carbon with desired properties towards effective VOSCs adsorption, post chemical modification/functionalization of commercially available activated carbon have been reported. Vega et al. (2013) identified that the oxidation of activated carbon with 6 N $HNO_3$ (wt./v, 1:10) for 20 min enriched the oxygen functional groups (e.g., -OH) in the carbon which interacted with the sulfur groups in ethyl mercaptan and dimethyl sulphide via hydrogen bonding and hence enhanced the removal of these species. Liu et al. (2017) investigated the production of N-dopped activated carbon via a 6-h post treatment on a commercial carbon, which involves oxidation with 30% $HNO_3$ at 60 ℃ for 30 min, followed by impregnation with melamine (wt/wt, 3:2, 25 ℃, 5 h), and finally pyrolyzing the mixture at 950 ℃ for 30 min. The carbon obtained showed increased adsorption capacity towards methyl mecaptan due to the enrichment of pyridinic and quaternary nitrogen. Jana and Sarkar (2018) reported that impregnation of activated carbon with 6.6 M $NH_4OH$ solution (wt./v, 1:8, 70 °C, 2 h, followed by placing at 35 °C for 24 h) improved the uptake of VOCSs. This was attributed to the introduction of basic N groups (e.g., nitrile, amino, cyclic amides and pyrrole like structure) and increased surface area and pore volume. Unfortunately, these post-treatment steps involve extensive prolonged processing times and are not cost-effective, and some negative impacts such as decreased surface area and porosity as a consequence of pore destruction from post chemical treatment processes may occur (Kupila et al., 2021; Wang et al., 2019), which are detrimental to adsorption. Therefore, it is worthwhile to consider a single-step process that combines activation and functionalization by impregnating the biomass with mixed $H_3PO_4$ and $HNO_3$ prior to carbonization, in order to design a carbon material with the desired textural properties and enriched with desired functional groups vital for improved VOSCs adsorption performance. Ultrasonic irridation has been reported to enhance the acid penetration into the biomass and cause structural deformation of lignocellulose biomass, and increase the nitration and oxidation of cellulose and lignin (Flores et al., 2021; Santos et al., 2021). We therefore hypothesized that a ultrasonic-assisted one step preparation of a biomass-based carbon sorbent will possess the desired physical and chemical properties for the removal of cat urine odor.

Sugarcane bagasse as a typical biomass waste in sugar production, nearly 33 million tons of sugarcane bagasse are produced in China every year (Chen et al., 2021). Although it is used for the generation of power for the sugar factories, there is still a significant amount of surplus which can be explored for value-adding utilization. Therefore, in the present work, a facile single-step preparation of a functionalized carbon material for the removal of MMB was investigated by impregnating sugarcane bagasse with $HNO_3$ and $H_3PO_4$, with the assistance of ultrasonic irridation. The pretreated biomass was dried, followed by a short-term smouldering pyrolysis in a muffle furnace to produce the carbon material. The textural properties and the surface functional groups of the carbons, with and without ultrasonication were determined, and the results were related to their MMB adsorption performance. The adsorption isotherms and kinetics of the selected adsorbent were evaluated, and density functional theory (DFT) calculations were performed to determine the extent of the interactions between the functional groups and the MMB molecules. The selected adsorbent was formulated into a sugarcane bagasse-based odor removal cat litter (SB-CL), of which excellent odor removal capacity was demonstrated.

## 2. Materials and methods

### 2.1. Materials

Sugarcane bagasse was provided by Guangxi Sugar Industry Group, China. $HNO_3$ (AR, ≥65% in $H_2O$) and $H_3PO_4$ (AR, ≥85% in $H_2O$) were purchased from Sinopharm Chemical Reagent Co., Ltd. 3-mercapto-3-methyl-1-butanol (MMB, 98%purity) was purchased from Shanghai Macklin Biochemical Co., Ltd.

### 2.2. Preparation of carbon from sugarcane bagasse

Sugarcane bagasse was milled with a lab-scale mill (Pulverisette 15, Fritsch, Germany), and was sized by sieving through a 30-mesh sieve. The bagasse that passed through the sieve was used both in preparing the carbon and for the cat litter formulation. Three acid treatments on bagasse were applied prior to pyrolysis process, and involve the following steps: (1) impregnate bagasse with 25 wt% $H_3PO_4$ at a mass ratio of 1:4, B-P; (2) impregnate sugarcane bagasse with 25 wt% $H_3PO_4$ and 15 wt% $HNO_3$ acids at a mass ratio of 1:2:2, B-NP; and (3) impregnate sugarcane bagasse with 25 wt% $H_3PO_4$ and15 wt% $HNO_3$ (1:2:2 mass ratio), followed by ultrasonic irridation for 30 min by using a ultrasonic probe (JY92-IIN, 20–25 KHz, 650 W), B-SNP. Each of the mixture were kept at ambient temperature for 24 h (acid treated bagasse namely B-P, B-NP, B-SNP were derived after drying at 45 ℃), followed by drying in 120 °C oven for 24 h. Thereafter, the dried materials were pyrolyzed at 700 °C for 30 min with a muffle furnace (KSW-8D-13) to produce the corresponding carbons, namely, D-P, D-NP, D-SNP. The obtained carbon samples were washed with deionized water to completely remove the unreacted acids and were then dried at 105 °C for 12 h. The carbon materials derived from (1)~(3) treatment procedures were denoted as AC-P, AC-NP and AC-SNP, respectively.

### 2.3. Characterization of carbon materials

The morphologies and the elemental compositions of the samples were determined by Field emission scanning electron microscope (SEM) and EDX spectroscopy with a ZEISS Sigma 300 device, Germany. High-resolution transmission electron microscopy was performed with HR-TEM, JEM- F200_TFEG under 200 kV acceleration voltage. XRD profiles of the samples were measured using an X-ray powder diffractometer (Rigaku D/MAX 2500 V, Japan) with Cu $K_\alpha$ radiation by a step-scanning method in the range of $2\theta = 10–80°$. Raman spectroscopy was used to measure the degree of graphitization of the carbons using (inVia Reflex, England). Cross-polarized magic angle spinning (CPMAS) $^{13}C$ nuclear magnetic resonance (NMR) was used to characterize AC-SNP on Bruker Avance III Spectrometer with 300 MHz magnet and a double air bearing

MAS probe. The surface area and porosity of the carbon materials were determined by nitrogen adsorption/desorption and BET method using ASAP 2020 Surface area and porosity analyzer (Micromeritics, USA) at 77.3 K. Prior to the adsorption experiment, samples were degassed at 300 °C under vacuum for 6 h, the Bruanauer-Ermmett-Teller (BET) method was used to calculate the specific surface area and Barret-Joyner-Halenda (BJH) model was used to generate the pore size distribution curves. The surface functional groups of the carbon materials were assessed by using a Fourier transformed infrared spectrometer (FTIR, IRTracer-100, Shimadzu, Japan) in the range 400–4000 $cm^{-1}$. The surface composition of the carbon materials were analyzed using the X-ray photoelectron spectrometry (XPS) technique (K-Alpha, Thermo Fisher Scientific, USA).

### 2.4. Mercapto-3-methyl-1-butanol (MMB) adsorption test

#### 2.4.1. Batch adsorption test

The adsorption tests were conducted by immersing the three carbon materials (AC-P, AC-NP, AC-SNP) in MMB solution. This concentration of 1.09 mg·mL$^{-1}$ in distilled $H_2O$ and a contact time of 100 min were selected to discriminate the adsorption capacities of the carbons. To monitor the odor removal effectiveness of the carbon materials, the MMB headspace concentration were determined based on the method adopted from Vaughn et al. (2020) with slight modifications. Detailed information is provided in Supplementary 1.1.

The percentage removal of MMB is described as follows:

$$R = \frac{C_0 - C_t}{C_0} \times 100\% \quad (1)$$

Where, $C_0$ (mg/mL) and $C_t$ (mg/mL) are the concentrations of MMB at initial, time $t$, respectively. $R$ is the removal percentage of MMB by carbon materials.

The best functionalized carbon material with the highest removal of MMB was selected for further adsorption experiments and for the fabrication of sugarcane bagasse-based cat litter formulations (SB_CLs).

The adsorption equilibrium of the selected functionalized carbon material was evaluated at different initial concentrations of 0.56, 0.79, 1.09, 1.50 and 3.57 mg/mL for 180 min. The MMB adsorption capacity at time t ($q_t$) and the equilibrium capacity ($q_e$) of the MMB were calculated by the following equations:

$$q_t = \frac{(C_0 - C_t)\cdot V}{m} \quad (2)$$

$$q_e = \frac{(C_0 - C_e)\cdot V}{m} \quad (3)$$

where $q_t$ (mg/g) and $q_e$ (mg/g) are the adsorption capacities of MMB on carbon materials at time $t$ and equilibrium, respectively. $C_0$ (mg/mL), $C_t$ (mg/mL), and $C_e$ (mg/mL) are the concentrations of MMB at initial, time $t$, and equilibrium, respectively. $V$ (L) is the volume of MMB solution. $m$ (g) is the adsorbent weight.

#### 2.4.2. MMB Adsorption isotherms and kinetics

To understand the surface properties and the affinity between the selected cabon, AC-SNP and MMB, the adsorption isotherm curve was acquired and fitted with Langmuir, Freundlich and Temkin isotherm models, which have been widely used to describe the adsorption of gaseous and liquid adsorbates over porous adsorbents (Guo et al., 2020; Gao et al., 2021; Yadav et al., 2021). Langmuir, Freundlich and Temkin isotherm models could be represented by the following equations:

$$q_e = \frac{q_m K_L C_e}{1 + K_L C_e} \quad (4)$$

$$q_e = K_F C_e^{\frac{1}{m}} \quad (5)$$

$$q_e = B\ln K_T + B\ln C_e \quad (6)$$

$$B = \frac{RT}{b} \quad (7)$$

Where, $q_e$ and $q_m$ (mg·g$^{-1}$) are the MMB adsorption capacity at equilibrium and the maximum adsorption quantity (mg·g$^{-1}$), $C_e$ is the concentration at equilibrium (mg·L$^{-1}$). $K_L$ is the Langmuir adsorption constant (in L$^{-1}$mg), $K_F$ is the Freundlich adsorption constant (mg·g$^{-1}$) and m is the heterogeneity factor or an empirical parameter dealing with the adsorption intensity. B is a constant related to the adsorption heat (J·mol$^{-1}$), $K_T$ is the equilibrium binding constant (L·g$^{-1}$) related to the maximal binding energy, $R$ (8.314 J·mol$^{-1}$ K$^{-1}$) is the universal gas constant, T (K) is the temperature, b is Temkin isotherm constant.

Isotherm studies were conducted at ambient temperature only in this study as the elevated temperature have impacts on the volatility of MMB which affect the MMB quantification from the headspace.

To investigate the mechanisms of MMB adsorption onto AC-SNP and identify the rate limiting step, adsorption kinetics were explored by analyzing the dynamic experimental data using three kinetic models including Pseudo-first-order (PFO), Pseudo-second-order (PSO) model and intra-particle diffusion model (IDP) (Yadav et al., 2021). The forms of the PFO, PSO and IPD models are described in the following Eqs. (8)–(10):

$$q_t = q_e(1 - \exp^{-k_1 t}) \quad (8)$$

$$q_t = \frac{q_e^2 k_2 t}{1 + q_e k_2 t} \quad (9)$$

$$q_t = k_{id}\sqrt{t} + C \quad (10)$$

Where, $q_t$ and $q_e$ are the amount of adsorbate adsorbed (mg/g) at any time t and equilibrium time (min). $k_1$ (min$^{-1}$), $k_2$ (g·mg$^{-1}$·min$^{-1}$) are the rate constant of PFO and PSO model, $k_{id}$ is the IPD model rate constant (g·mg$^{-1}$·min$^{-1}$) and C (mg/g) is the intercept, which is a constant related to the thickness of the boundary layer.

#### 2.4.3. Error function analysis

The fitting of the isotherms and kinetic models for the experimental data was evaluated by statistical parameters. The $R^2$ (or $Adj.R^2$) was used to evaluate the validity of the models and was calculated based on Eq. (11). $R^2$ reflects the goodness-of-fit (model accuracy) of the isotherm kinetic model to the experimental data. The Adjusted $R^2$ also indicates how well the data fits with a model, but considers the number of terms in a model.

$$Adj.\quad R^2 = 1 - \frac{\sum (q_{exp} - q_{cal})^2}{\sum (q_{exp} - q_{mean})^2} \times \frac{N-1}{N-p} \quad (11)$$

Where $q_{cal}$ and $q_{exp}$ (mg/g) are the adsorption capacities from model calculation and experiment, respectively. $q_{mean}$ (mg/g) is the average experimental adsorption capacity, $N$ is the number of experimental data, and $p$ is the number of model parameters.

### 2.5. Density functional theory (DFT) calculations

The interactions between the MMB molecules and the functional groups of the carbon material were investigated with Gaussian 16 software. Detailed information please refer to Supplementary 1.2. To evaluate the interaction of MMB molecule with AC-SNP, adsorption energy ($E_{ad}$) was calculated as follows:

$$E_{ad} = E_{(M-A)} - E_M - E_A \quad (12)$$

where $E_{(M-A)}$ is the total energy of MMB molecule adsorbed on the AC-SNP surface, and $E_M$ and $E_A$ are the total energies of the MMB

molecule and AC-SNP, respectively. $E_{ad} < 0$ can be attributed to effective adsorption of MMB by adsorbent.

### 2.6. In vitro cytotoxicity assay and the odor removal cat litter formulation

In vitro cytotoxicity tests for AC-SNP were conducted by using CCK-8 assay with GES-1 cell (Guo et al., 2021). Detailed procedures are provided in the Supplementary 1.3.

The AC-SNP was then used for the odor removal component for the sugarcane bagasse-based cat litter (SB_CL), which was formulated with bagasse, guar gum and modified lignin from bagasse (Supplementary 1.4). The properties of the optimized SB_CL formulation was compared with two commercially available cat litters i.e., Tofu cat litter and bentonite clay litter.

## 3. Results and discussion

### 3.1. Characterization of activated carbon materials

#### 3.1.1. Textural properties of the carbon materials

*3.1.1.1. Morphology analysis.* The morphologis of AC-P, AC-NP and AC-SNP are shown in the SEM micrographs in Fig. 1. AC-P exhibits fluffy bulky shape and heterogeneous surface topography (Fig. 1b), and the high magnification SEM image reveals the carbon to contain many small cavities of the porous structure. The formation of the porous structure is due to activation with $H_3PO_4$, which is known to promote the dehydration of lignocellulosics especially at the edges of the cell walls, and also enhances the formation of crosslinked oxygen-containing functional groups (Guo et al., 2020). During the pyrolysis process, with $H_3PO_4$ volatilization at higher temperatures there were water vapor formation and inhibition of tar formation, both of which resulted in the formation of porous structures (Guo et al., 2020). While AC-NP shows a similar bulky shape as AC-P, a denser surface structure was formed as shown in the image in Fig. 1(c). The surface topography was uneven with relatively smaller pore size than AC-P, and may be because of the decrease in proportion of $H_3PO_4$ due to the addition of $HNO_3$. Nonetheless, as stated in an earlier study, $HNO_3$ activation increased the microporosity of carbonaceous-derived carbons (Li et al., 2018). AC-SNP as shown in Fig. 1(d), also present fluffy and porous structure, though the pore size is larger than that of AC-NP. This may be due to enhanced etching of the cell walls via ultrasonication (Chen et al., 2020), which further impacted on the carbon structure formed (Yu et al., 2019). The morphology of sugarcane bagasse (B), $H_3PO_4$ treated bagasse (B-P), mixed $H_3PO_4/HNO_3$ treated bagasse (B-NP) and ultrasonic-assisted mixed $H_3PO_4/HNO_3$ treated bagasse (B-SNP) were also analysed and are shown in Fig. 1a (1)-(4), respectively. Cell deformation were observed for all pretreated samples. However, for B-SNP (Fig. 1a (4)), there was size and crystallinity reduction, and macrostructural reorganization due to the interaction between the lignocellulose biomass and the ultrasonic waves which weakened the intermolecular hydrogen bonding among the lignocelluose components (Flores et al., 2021).

The morphology of AC-SNP was also investigated by HR-TEM. According to the TEM image shown in Fig. 1(e), porous amorphous carbon with highly disordered and randomly curled structure are observed, and the SAED analysis shows diffuse diffraction rings, which indicates no identifiable crystalline symmetry (Zeng et al., 2017).

*3.1.1.2. Pore structure of the carbon materials.* The porosity characteristics of the three carbon samples were investigated by measuring the $N_2$ adsorption-desorption isotherms at 77 K. The results are shown in Fig. 2 (a), and the corresponding pore size distribution of the carbons are shown in Fig. 2(b). Both AC-P and AC-SNP exhibit Type IV isotherms with $H_4$ type of hysteresis loop (Üner et al., 2018), which could be attributed to the presence of both mesopores (2–50 nm) and micropores (< 2.0 nm) (Jiang et al., 2017; Mousavi-Qeydari et al., 2021), and the capillary condensation occurred in the mesopores. A relatively higher adsorption of $N_2$ were found with AC-SNP than AC-P because of increased specific surface area (1603.15 $m^2/g$) and pore volume (1.39 $m^3/g$) as illustrated in Table 1. AC-NP shows a Type I isotherms, the major uptake occurs at a relatively low pressure, which indicates the existence of highly microporous structure confirming the SEM

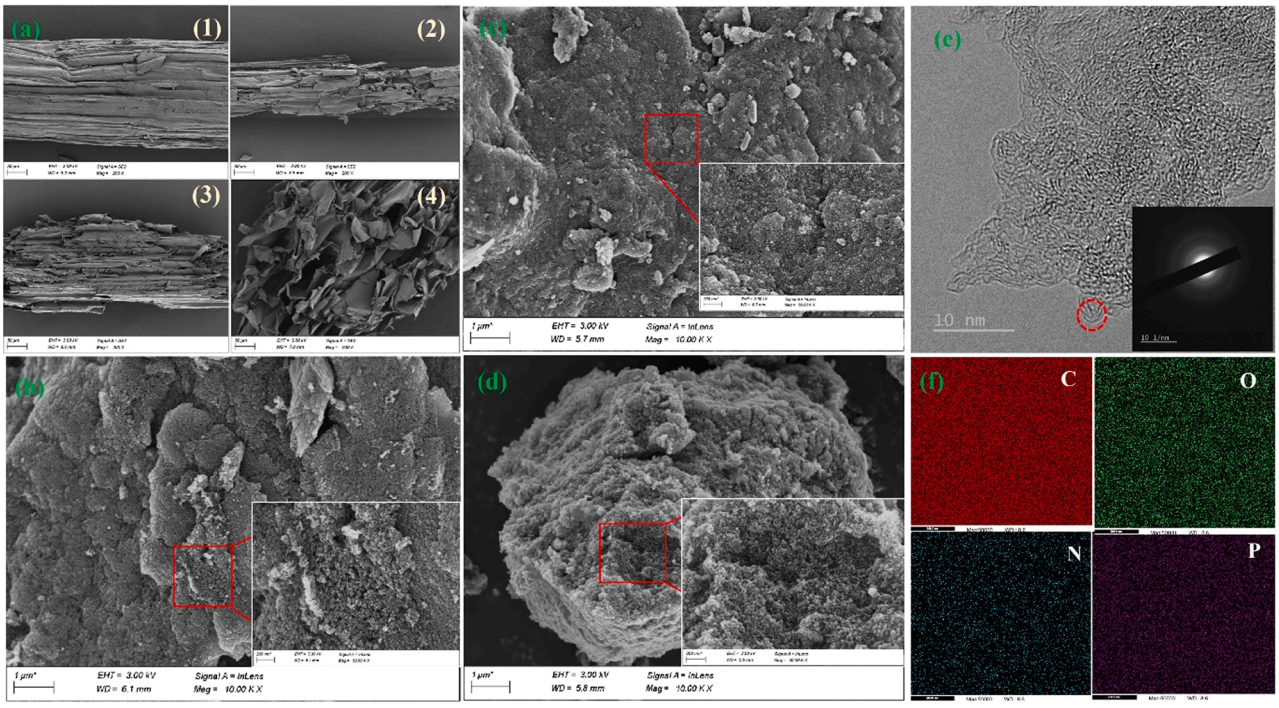



**Fig. 1.** SEM images of: (a-1) bagasse, (a-2) $H_3PO_4$ treated bagasse, (a-3) mixed $H_3PO_4/HNO_3$ treated bagasse, (a-4) ultrasonic-assisted $H_3PO_4/HNO_3$ treated bagasse, and the carbon samples derived (b) AC-P, (c) AC-NP, (d) AC-SNP, and HRTEM image and SAED (e) and C, N, O, P elemental mapping (f) of AC-SNP.

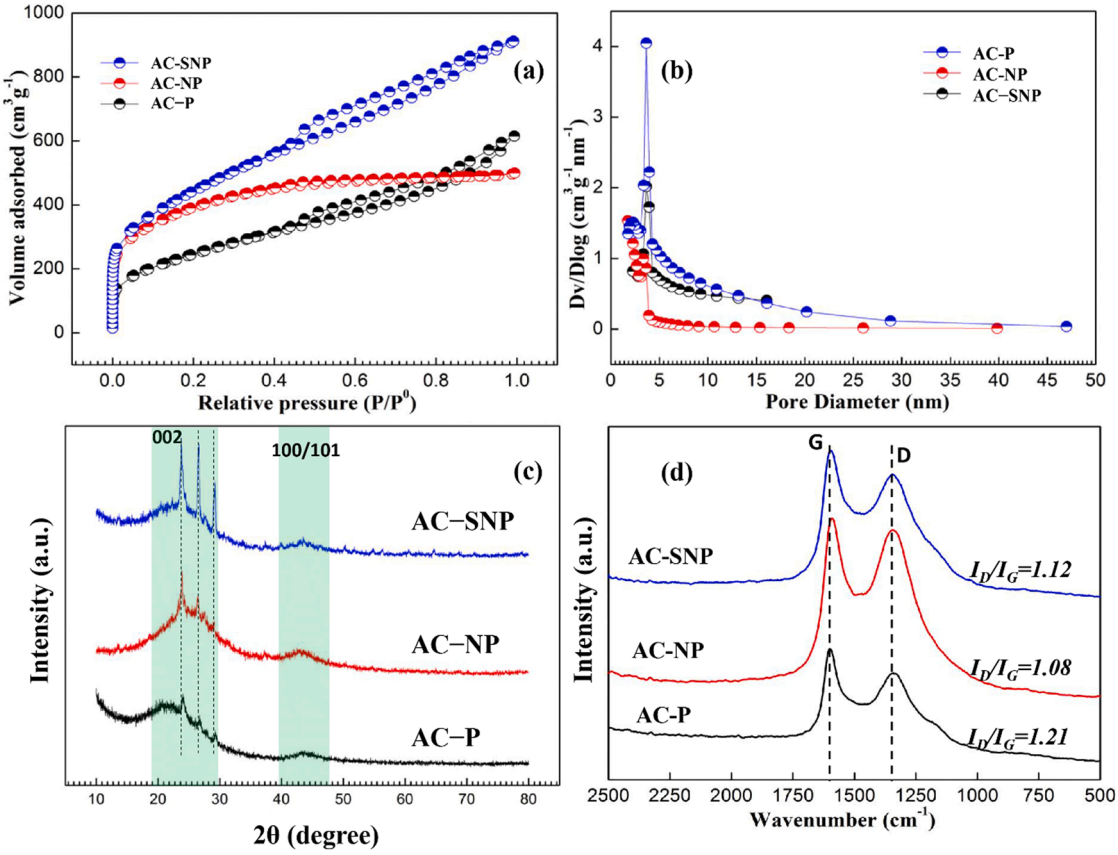



**Fig. 2.** (a) $N_2$ adsorption-desorption isotherm at 77.3 K, (b) differential pore size distribution, (c) Raman spectra, and (d) XRD spectra of AC-P, AC-NP and AC-SNP.

**Table 1**
Textural properties calculated from nitrogen adsorption desorption test at 77.3 K.

| Carbon materials | $S_{BET}$ ($m^2$ $g^{-1}$) | $V_{total}$ ($cm^3$ $g^{-1}$) | $V_{micro}$ ($cm^3$ $g^{-1}$) | $V_{meso}$ ($cm^3$ $g^{-1}$) | $d_p$ (nm) |
|---|---|---|---|---|---|
| **AC–P** | 898.63 | 0.92 | 0.12 | 0.80 | 4.50 |
| **AC–NP** | 1414.58 | 0.76 | 0.17 | 0.59 | 2.16 |
| **AC–SNP** | 1603.15 | 1.39 | 0.03 | 1.35 | 3.46 |

micrograph. A $H_4$ type hysteresis loop was also identified, indicating the co-existence of mesopores, however, at significantly lower intensity than the micropores (Fig. 2b). The increased porosity of AC-NP compared to AC-P is mainly due to the addition of $HNO_3$, which improved the exfoliation of the lignocellulosic fractions (Fig. 1a 3) and their hydrolysis, in particular the hemicellulose fraction as evidenced by the increased arabinose, galactose and xylose content in the hydrolysate (Table S1) (Ning et al., 2020). When ultrasonication was applied, there was further degradation enhanced by the increased penetration and accessibility of the mixed acids to the lignocellulosic fractions arising from both physical (size reduction) and sonochemical (radicals e.g., •H, •OH formation) effects of ultrasonic irridation (Flores et al., 2021). A further structure deformation (Fig. 1a 4) is associated with increased hemicellulose hydrolysis (Table S1), and accounted for the highest porosity and specific surface area obtained with AC-SNP.

*3.1.1.3. XRD analysis.* The extent of the crystallinity of the carbon materials was further confirmed by XRD analysis, and the results are presented in Fig. 2(c). The XRD patterns of the three carbon materials display broad peaks at $2\theta = 20 \sim 30^\circ$ and 44°, which are assigned to the stacking peak of π-conjugated layers (002) and graphitic carbon diffractions (10X, overlayed 100 and 101 planes), respectively (Zhou et al., 2019; Deshan et al., 2021). This represents amorphous carbon structures with the presence of microcrystalline carbon fragments randomly distributed. The results given by XRD are in good agreement with that of TEM results in terms of the amorphous nature of the carbons. The 002 peak shifted towards the larger angle (from 22.5° for AC-P to 25° for AC-NP, and 23.6° for AC-SNP), indicating the increase in the stacking thickness (Vazquez-Santos et al., 2012) with the use of mixed acids, while the additional treatment with the ultrasonication slightly reduced stacking. Three sharp peaks at 2θ of 23.6° (0.38 nm), 26.4° (0.34 nm) and 29.1° (0.31 nm) were identified for all the three carbon samples, which could be indexed as turbostratic carbon crystallites, crystallographic structure of quartz and silicon oxide hydrate, respectively (Gupta et al., 2019; Saafie et al., 2019). There was increased intensity with AC-NP and AC-SNP compared to AC-P, may be related to the increased dissolution of carbohydrates e.g., monosaccharides from bagasse during pretreatment as shown in Table S1, thereby increasing the percentage proportion of silica, as well as allowing ultrastructural changes to occur during carbonization.

*3.1.1.4. Raman analysis.* Raman spectroscopy was further used to probe the impacts of the pretreatment on the microstructure of the carbon materials (Fig. 2d). The signals in Raman spectra are arising from the lattice vibration and are very sensitive to structure disorder. Two major characteristic peaks i.e., the D band (near 1350 $cm^{-1}$) and the G band (near 1580 $cm^{-1}$) were identified, where the D band represents for the disturbed graphitic lattice and are indicative of functionalities in the carbon matrix, while the G band refers to the graphene/graphene-like sheets (Adio et al., 2020; Saini et al., 2021). The intensity ratio of $I_D/I_G$ for AC-P was the highest (1.21), implying the most disordered structure, followed by AC-SNP (1.12) and AC-NP (1.08), suggesting using mixed acids as activating agents resulted in carbon with decreased structural disorder, and the introduction of ultrasonication in the

impregnation process further destroyed the oriented structures.

### 3.1.2. Surface functional properties of the carbon materials

*3.1.2.1. Elemental composition of carbon materials and the precusors.* To understand the evolution of elemental composition in the precursors and carbon materials, raw bagasse, acid pretreated bagasse, oven dried bagasse/acids mixture, and carbon samples were analysed with EDAX mapping (Table 2). Raw bagasse is composed of 53.42% carbon, 42.63% oxygen, 2.83% nitrogen, and 1.12% phosphrous. A measurement of the proportions of O and N contents in the materials are the O/C and N/C ratios. Decreased O/C ratio was found for B-P compared with raw bagasse and may be due to the hydrolysis and cleavage of the C-O bonds of hemicellulose when $H_3PO_4$ was added (Collard et al., 2014). Increased N/C and O/C ratios were observed in B-NP and B-SNP are due to the nitration of the lignin fraction and the oxidation of carbohydrates occurring simultaneously with the addition of $HNO_3$ (Ning et al., 2020), This is supported by the increased intensities of -COOH and -N-O groups at 1741, 1536 and 1330 $cm^{-1}$ in the FTIR spectrum (Fig. S 2a). The slightly higher N/C ratio with B-SNP could be due to the enhanced oxidation and nitration arising from the ultrasonic treatment which is associated with both physical (acoustic cavitation, shock waves and acoustic streaming) and chemical (radicals' formation) effects. A recent study has shown that ultrasonication does facilitate the nitration of cellulose at low $HNO_3$ concentration (Santos et al., 2021).

After acid impregnation, the mixture was further dried at 120 ºC for 24 h. Significant decrease in the O/C ratio were identified with all three dried samples (D-SNP > D-NP > D-P), which is likely due to further acid catalyzed dehydration and rearrangement reactions of the lignocellulosic fractions at the elevated temperature and gradual increase in the acid concentration during the drying process (Long et al., 2017). This result was confirmed by the FTIR spectrum (Fig. S 2b) in which there are significant reduction of the C-O stretching, H-C deformation, and H-C stretching signals at 2909, 1460 and 1044 $cm^{-1}$ (Cai et al., 2019; Rashid et al., 2018). A slight increase of N/C ratio (0.126) was found with D-SNP compared with that of D-NP (0.086), indicating that the nitrate groups in the ultrasonicated assisted acid treated bagasse is more stable towards heating and the degradation process. So, this is a beneficial approach to enrich N sites on the surface of carbon prepared in-situ as reported in an earlier study by Graglia et al. (2016).

The oven dried mixture was further pyrolyzed at 700 ºC for 30 min where further dehydration, depolymerization, fragmentation and molecule structure rearrangement occur and char formation (Collard and Blin, 2014). Char forms at elevated temperatures (500–700 ºC) with the formation of benzene rings and the combination of these rings into a polycyclic structure and the release of water and incondensable gases. As shown in Fig. S3, the $^{13}C$ NMR spectrum of AC-SNP showed an intense peak at 125 ppm, which corresponds to the aromatic structure confirming the results (McGrath et al., 2003). Besides, two peaks at 30 and 215 ppm are related to the aliphatic and carbonyl-ketone structures (McGrath et al., 2003). Despite similar pyrolysis conditions, the biomass constituents and the interactions between different fractions arising from pretreatment, have impacts on the final carbon yields and carbon properties (Collard and Blin, 2014). The N/C and O/C ratio of the carbon prepared was AC-SNP > AC-NP > AC-P. The absence of N in AC-P indicated it is originated from the $HNO_3$. The EDX mappings of AC-SNP are shown in Fig. 1(f), with uniform distribution of C, N, O and P is an confirmation that the phosphorylation and oxidation processes occurred uniformly in the entire carbon material (Ternero-Hidalgo et al., 2016).

**Table 2**
Elemental composition evolution in carbon and their precursors from EDAX mapping analysis.

| Materials | C K (%) | N K (%) | O K (%) | P K (%) | N/C ratio | O/C ratio |
|---|---|---|---|---|---|---|
| **Bagasse** | 53.42 | 2.83 | 42.63 | 1.12 | 0.05 | 0.80 |
| **B-P** | 56.57 | 3.16 | 38.25 | 2.03 | 0.06 | 0.68 |
| **B-NP** | 47.79 | 4.97 | 44.69 | 2.55 | 0.10 | 0.94 |
| **B-SNP** | 48.67 | 5.33 | 43.96 | 2.04 | 0.11 | 0.90 |
| **D-P** | 66.00 | 3.51 | 27.13 | 3.36 | 0.05 | 0.41 |
| **D-NP** | 69.12 | 5.97 | 23.75 | 1.16 | 0.09 | 0.34 |
| **D-SNP** | 61.17 | 7.69 | 30.09 | 1.05 | 0.13 | 0.49 |
| **AC-P** | 87.25 | 0 | 11.14 | 1.61 | 0.00 | 0.13 |
| **AC-NP** | 83.09 | 3.00 | 11.47 | 2.44 | 0.04 | 0.14 |
| **AC-SNP** | 81.27 | 4.04 | 13.48 | 1.24 | 0.05 | 0.17 |

*3.1.2.2. FT-IR spectroscopy analysis.* The evolution of the functional groups of the carbon materials derived from bagasse from different pretreatment processes were identified with Fourier transform Infrared (FT-IR) spectroscopy, the results are presented in Fig. 3(a). All three carbons show peaks at 3435, 1629 $cm^{-1}$, which could be assigned to stretching vibration of O-H in carboxyl and phenol (Gokce and Aktas, 2014; Jawad et al., 2017), the highly conjugated carbonyls (quinone groups) and C═C stretching vibration in the aromatic ring, respectively (Pezoti et al., 2016; Kalaiyarasan et al., 2019; Yadav et al., 2021). The weak absorption peaks observed at 2932–2972 $cm^{-1}$ from all three carbons are attributed to the aliphatic C-H stretching vibration (Guo et al., 2020). The bands located at 1532 and 1523 $cm^{-1}$ identified in AC-NP and AC-SNP are responsible for the stretching vibration of C-N═O or C═N groups (Li et al., 2019), further confirmed the N-containing groups were successfully preserved in the carbon matrix. The peak at 1191 $cm^{-1}$ may be also assigned to the stretching mode of hydrogen-bonded P═O of phosphates or polyphosphates, to O-C stretching vibrations in P-O-C (aromatic) linkage and to P═OOH were identified in all the three carbon materials, indicated that the $H_3PO_4$ acting as an acid catalyst to promote the bond cleavage reactions and the formation of cross-links via cyclization and condensation and to combine with organic species to form phosphate and polyphosphate bridges that connect and crosslink biomass (Sych et al., 2012; Yagmur et al., 2018), while with the assistance of ultrasonication, the peak shifted from 1191 $cm^{-1}$ to 1140 $cm^{-1}$, suggesting the ultrasonication enhanced the formation of phosphor-carbonaceous compounds (Sych et al., 2012). Peak at 1045 $cm^{-1}$ that related to the $P^{+}$-$O^{-}$ in acid phosphate esters and due to the symmetrical vibration in polyphosphate chain P-O-P (Sych et al., 2012) were found in both AC-P and AC-NP, while disappeared or significantly weakened in AC-SNP. While new peaks at 1750 and 1442 $cm^{-1}$ that corresponded to C═O stretching vibration of carboxylic acids or lactone and the C═C stretching in aromatic ring were observed in AC-SNP (Demiral et al., 2021; Md Salim et al., 2021), which indicates that the ultrasonication process promotes biomass oxidation (Mason, 1986) and endow the carbon with increased specific surface area and enriched surface-active sites (Ma et al., 2013).

*3.1.2.3. X-ray photoelectron spectroscopy (XPS) analysis.* XPS was used to further investigated the nature of functional groups and the bonding configurations. The survey spectra of the carbons are shown in Fig. 3(b). Table 3 detailed the elemental content of the outermost surface of the carbon materials, and similar to the results from the EDX elemental analysis, increased N1s, O1s and P2p were observed in AC-NP and AC-SNP compared with ACP, while the increment is more pronounced in the later except for the slightly lower N content. However, relatively higher O, N and P contents were observed than the results obtain from EDX mapping, indicating that the chemical activation process is more predominant at the outermost surface of the carbons than further inside the carbons (Ternero-Hidalgo et al., 2016). The increased P content in AC-NP and AC-SNP further confirmed that the phosphorylation process was enhanced with the use of $HNO_3$ and ultrasonication. The high resolution C1s spectrum (Fig. 3c) shows the presence of bond types of C-C/C═C, C-O/C-N, and C═O (Tehrani et al., 2019). The deconvoluted AC-SNP O1s peak (Fig. 3d) of AC-SNP shows that the strongest peak

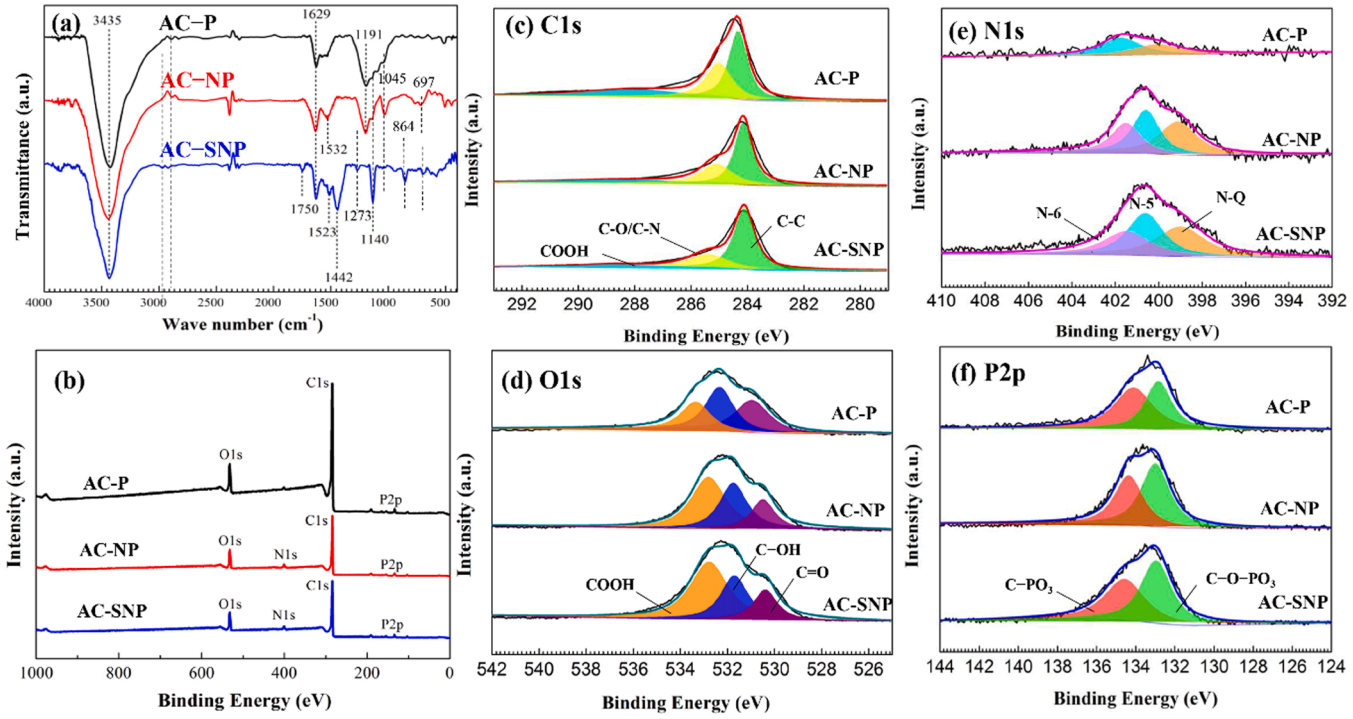



**Fig. 3.** (a) FTIR spectra, (b) XPS spectra, and high-resolution of (c) C1s XPS spectra, (d) O1s XPS spectra, (e) N1s XPS spectra and (f) P2p XPS spectra of AC-P, AC-NP and AC-SNP.

**Table 3**
Quantitative characterization of the surface elemental compositions of the as-prepared adsorbents through XPS.

| Carbon materials | Atomic (wt%) | | | | O1s (wt%) | | | N1s (wt%) | | | P2p (wt%) | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | C1s | P2p | O1s | N1s | C═O/ COOR | C-OH | COOH | N–6 (Pyridinic N) | N–5 (Pyrrolic N) | N–Q (Graphitic N) | $C\text{-}PO_3$/ C2-PO3 | $CO\text{-}PO_3$/ (C-O)3-PO |
| AC–P | 86.50 | 2.22 | 9.83 | 1.45 | 3.50 | 3.82 | 2.51 | 0.00 | 0.57 | 0.88 | 1.25 | 0.99 |
| AC–NP | 78.94 | 3.06 | 13.16 | 4.84 | 2.39 | 4.39 | 6.38 | 1.86 | 1.74 | 1.23 | 1.73 | 1.33 |
| AC–SNP | 77.31 | 3.31 | 14.70 | 4.68 | 2.54 | 4.29 | 7.87 | 1.67 | 1.77 | 1.24 | 1.78 | 1.53 |

located at 532.8 eV corresponds to the -COOH group (53.55%), which represents the surface acidity, followed by the peak at 531.7 eV (C-O of alcohol and/or ether) and 530.4 eV (C═O of ketone, carbonyl, lactone and/or anhydride) that reflect the surface basicity (Guo et al., 2020; Yao et al., 2020). The percentage of -COOH was lowest in AC-P (25.50%), suggesting that nitration increased the formation of -COOH proportion on the carbon surface which was further increased through the application of ultrasonication. The high resolution N1s spectrum (Fig. 3e) that fitted by three nitrogen species of, pyridinic-N (N-6), pyrrolic-N (N-5), and graphitic-N (G-N) centered at 399.0, 400.6 and 401.5 eV, respectively (Han et al., 2019; Gao et al., 2021) were identified on the surfaces of AC-NP and AC-SNP. Pyrrolic N was absent in AC-P, indicating the formation of N-5 is related to the nitration of bagasse. The deconvolution of P2p spectrum (Fig. 3f) reveals that $C\text{-}PO_3$/C2-$PO_3$ (132.8 eV) and C-O-$PO_3$/$(C\text{-}O)_3$-PO (134.1 eV) are the two forms of P present (Ternero-Hidalgo et al., 2016). Introduction of $HNO_3$ and ultrasonication were found to increase the contents of both $C\text{-}PO_3$/C2-$PO_3$ and C-O-$PO_3$/$(C\text{-}O)_3$-PO in the carbon produced.

### 3.2. MMB (3-mercapto-3-methyl-1-butanol) adsorption with carbon materials

#### 3.2.1. MMB adsorption and adsorption parameters optimization

The initial batch adsorption tests were carried out with AC-P, AC-NP and AC-SNP using 1.09 mg/mL of MMB at a contact time of 100 min to discriminate the adsorption performance between the three carbons prepared. As shown in Fig. 4(a), AC-SNP gave the highest adsorption capacity of 204.48 mg/g at 25 °C (94.02% removal efficiency of MMB), followed by AC-NP with a value of 191.48 mg/g (87.99% removal efficiency) and AC-P with a value of 132.88 mg/g (61.06% removal efficiency) in that order. A simple plot of the carbons surface areas and their removal efficiencies indicated a linear relationship ($R^2 = 0.99$, Fig. S4) an indication that surface area is the key parameter response for effective adsorption of MMB. On the basis of these results, AC-SNP was selected for further evaluation.

The adsorption of MMB by AC-SNP was studied under different conditions with the variation of several parameters, one at a time, while others were kept constant. The impact of the adsorbent dosage was investigated and results are shown in Fig. 4(b). As the dosage of carbon increased from 2.5 to 20 g/L, the removal of MMB increased from 66.25% to 99.21%. However, the removal efficiency of MMB increased sharply to up to 5 g/L (98.31%) then increased slightly to the adsorbent loading to 20 g/L (99.21%). This means that increasing the cabon loading pass 5 g/L did not improve the adsorption rate, and that most of the active site were already occupied. Therefore, the dosage of 5 g/L was selected for further tests.

The effect of contact time on the adsorption of MMB on AC-SNP was studied at several time intervals from 0 to 180 min Fig. 4(c) shows that MMB uptake increased with increasing contact time. This was particularly so in the early stages of the adsorption process, with 29.31% MMB adsorbed in the first 5 min, 50.94% of MMB adsorbed within 10 min, and 68.85% of MMB adsorbed within 20 min, implying that a majority

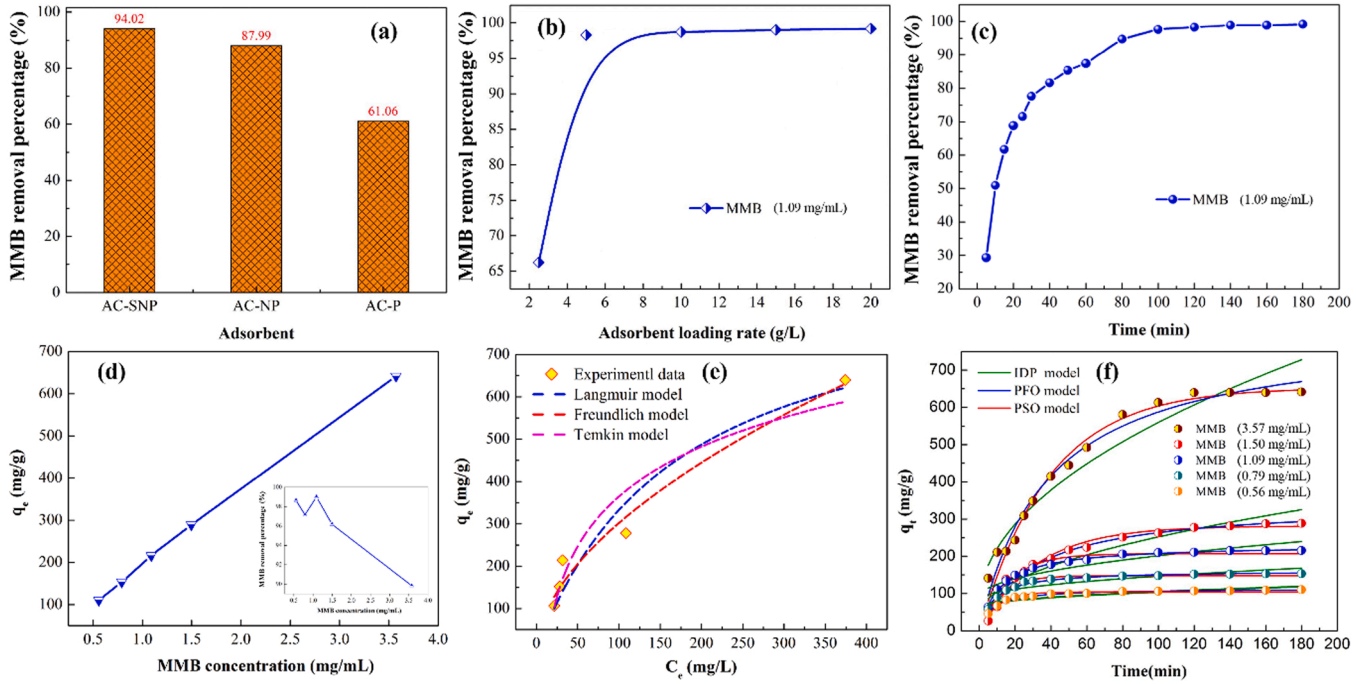



**Fig. 4.** Effect of (a) adsorbent type, (b) carbon loading ratio, and (c) adsorption time on the adsorption of MMB ($C_0$ = 1.09 mg·mL$^{-1}$); (d) Effect of the initial MMB concentration on the adsorption of MMB ($C_0$ = 0.56–3.57 mg·mL$^{-1}$, t = 100 min). (e) MMB adsorption isotherms of the AC-SNP at 298 K, and (f) and adsorption kinetics of MMB on AC-SNP.

of MMB (odor) could be removed in a relatively short time. The adsorption approached equilibrium at around 100 min with a removal percentage of 97.65%, as the removal increase to 180 min was small.

Fig. 4(d) shows the influence of the initial MMB concentration on the adsorption capacity of AC-SNP. With the increase of MMB concentration from 0.56 to 3.57 mg/mL, the overall removal percentage showed a decline trend from 98.64% to 89.84% except for the initial concentration of 1.09 mg/mL were the highest removal percentage of 99.00% (inset figure in Fig. 4d) was recorded. The equilibrium adsorption capacity increased from 110.48 to 641.44 mg/g (Fig. 4f), due to increased adsorption rate of the available active sites with the adsorbate concentration of 1.09 mg/mL (Yadav et al., 2021). Therefore, the concentration of 1.09 mg/mL was considered as the optimum adsorption concentration.

### 3.2.2. *Adsorption isotherms and kinetics*

Adsorption isotherms show the relationship between the residual concentration of MMB in aqueous solution and the equilibrium adsorption capacity of MMB at 25 °C (Fig. 4d). The adsorption isotherms were analyzed using the Langmuir, Freundlich, and Temkin models as shown in Eqs. (4) - (6). The Langmuir isotherm model assumes a monolayer adsorption process on an even adsorbent surface, while Freundlich isotherm model presumes that the adsorption of MMB takes place by multilayer adsorption process between the adsorbed molecules. Temkin model considers the heat of adsorption associated with the interactions between adsorbate and the adsorbent, and presumes the adsorption heat drops linearly as the adsorption rises (Jawad et al., 2019; Reghioua et al., 2021; Yadav et al., 2021). The results of Langmuir, Freundlich and Temkin models fittings to the experimental data are presented in Fig. 4(e), and the correlation coefficients (*Adj*.$R^2$) were determined to evaluate the best fit, which reflects the applicability of the models to the experimental data. As shown in Table 4, the obtained *Adj*. $R^2$ of the Freundlich model (0.96) is higher than that of Langmuir model (0.91) and Temkin model (0.89), revealing that Freundlich model suits better to the data of MMB adsorption by AC-SNP than the Langmuir and Temkin models. This implies that the adsorption of MMB to AC-SNP is a heterogeneous multi-layer adsorption process (Reghioua et al., 2021). The constant *m* represents the adsorption intensity or surface heterogeneity. A value of 0.56 was obtained for 1/*m* in this study, indicating that the adsorption of MMB by AC-SNP was favorable (Guo et al., 2021). The relatively larger total pore volume and mesoporousity of AC-SNP than AC-NP (Table 1) will allow MMB molecules at higher concentrations to enter the mutli-layers. As the *Adj*.$R^2$ (0.91) for Langmuir model was > 0.9, means the Langmuir model could also fit well with the equilibrium data, and the $R_L$ of the model was 0.046–0.24 at the tested initial concentrations, indicating the adsorption is favorable. The high affinity between the MMB and AC-SNP indicates that the chemisorption is the predominant adsorption mechanism.

**Table 4**
Adsorption isothermal model parameters.

| Isotherm models | Isotherm parameters | |
|---|---|---|
| **Langmuir** | $q_m$ *(mg/g)* | 908.69 |
| | $K_L$ *(L/mg)* | 0.01 |
| | $R_L$ | 0.05–0.24 |
| | *Adj.* $R^2$ | 0.91 |
| **Freundlich** | $K_F$ | 23.19 |
| | $m^{-1}$ | 0.56 |
| | *Adj.* $R^2$ | 0.96 |
| **Temkin** | $B_1$ | 169.58 |
| | $K_T$ | 0.09 |
| | *Adj.* $R^2$ | 0.89 |

Kinetics models of PFO, PSO and IPD were used to examine the adsorption mechanism and the respective rate controlling step, including mass transport and chemical reaction. The fittings of the experimental data with the three kinetic models for the adsorption of MMB at ambient temperature are shown in Fig. 4(f). All kinetic parameters including the adjusted correlation coefficients (*Adj.* $R^2$) are listed in Table 5. The best fitting kinetic model was evaluated based on the *Adj.* $R^2$ value. For the PFO fitting lines for MMB regression, the *Adj.* $R^2$ ranged from 0.95 to 0.98, while that for the PSO fitting lines for the MMB adsorption data ranged from 0.98 to 0.99, which means that the

**Table 5**
Adsorption kinetic model parameters.

| Kinetic models | Parameters | Initial MMB concentration $C_0$ (mg/mL) | | | | |
|---|---|---|---|---|---|---|
| | | 0.56 | 0.79 | 1.09 | 1.50 | 3.57 |
| PFO model | $q_e$ (mg/g) | 104.3000 | 147.9360 | 207.1239 | 281.4711 | 653.1465 |
| | $k_1$ ($min^{-1}$) | 0.1009 | 0.0878 | 0.0630 | 0.0318 | 0.0258 |
| | *Adj.* $R^2$ | 0.9476 | 0.9734 | 0.9561 | 0.9755 | 0.9770 |
| PSO model | $q_e$ (mg/g) | 113.2738 | 161.6492 | 231.5893 | 341.2375 | 810.3264 |
| | $k_2$ ($g \cdot mg^{-1} \cdot min^{-1}$) | 0.0014 | 0.0008 | 0.0004 | 0.0001 | 0.0000 |
| | *Adj.* $R^2$ | 0.9841 | 0.9851 | 0.9949 | 0.9806 | 0.9789 |
| IPD model | $k_{id}$ ($g \cdot mg^{-1} \cdot min^{-1}$) | 4.1320 | 6.3021 | 11.1717 | 21.4072 | 49.3430 |
| | C | 63.2435 | 83.5534 | 89.8317 | 38.1054 | 66.0520 |
| | *Adj.* $R^2$ | 0.6797 | 0.6698 | 0.7945 | 0.8800 | 0.9339 |

PSO model describes the adsorption process of MMB onto AC-SNP better than the PFO model, confirming that the adsorption process is dominated by chemisorption rather than physical adsorption. Chemisorption is the rate determining step that limits the adsorption process (Jawad et al., 2020; Guo et al., 2021; Yadav et al., 2021). The IPD model was also used to fit the adsorption data, and the *Adj.* $R^2$ were 0.67–0.93 at different initial concentrations, which are lower than those of both PFO and PSO models. Additionally, the *Adj.* $R^2$ values increased with increasing MMB initial concentration, implying good correlation of the adsorption data at higher MMB initial concentrations, which indicated that the intra-particle diffusion plays a role in the rate limiting step with increased initial MMB concentration (Yadav et al., 2021). The inference from this is that at relative higher MMB concentration, two or more processes are involved in the adsorption process, which include external diffusion and intra-particle diffusion, with the later exhibiting a lower diffusion rate. In summary, comparing the results from the three kinetic models, PSO is the predominant process, indicating that chemisorption is the rate limiting step of the adsorption process (Bedin et al., 2018).

### 3.3. DFT calculations

#### 3.3.1. Carbon and MMB models, and molecular surface electrostatic potential (ESP)

To understand the extent of the interactions between the functional groups on carbon surface and the MMB molecules, a systematic study was carried out using DFT calculations. The carbon structure with 20 benzene rings is the basic structure used in this study, with the functional groups (Fig. 3) of -COOH, C═O, C-OH, C-O-C (including bridging C-O-C), C-$PO_3$, C-O-$PO_3$, N-6, N-5 and G-N present on the carbon surface. The MMB molecules and eleven surface models were geometry optimized by the DFT-B3LYP function, and the optimized model structures are shown in Fig. S5 (a-l). The surface electrostatic potential (ESP) of the functionalized carbon materials and MMB was analysed as shown in Fig. S6 (a-l) and interpreted in Supplementary 2.1. Based on the results obtained with the functional groups on the AC-SNP and MMB, it can be inferred that good interactions between the adsorbent and adsorbate.

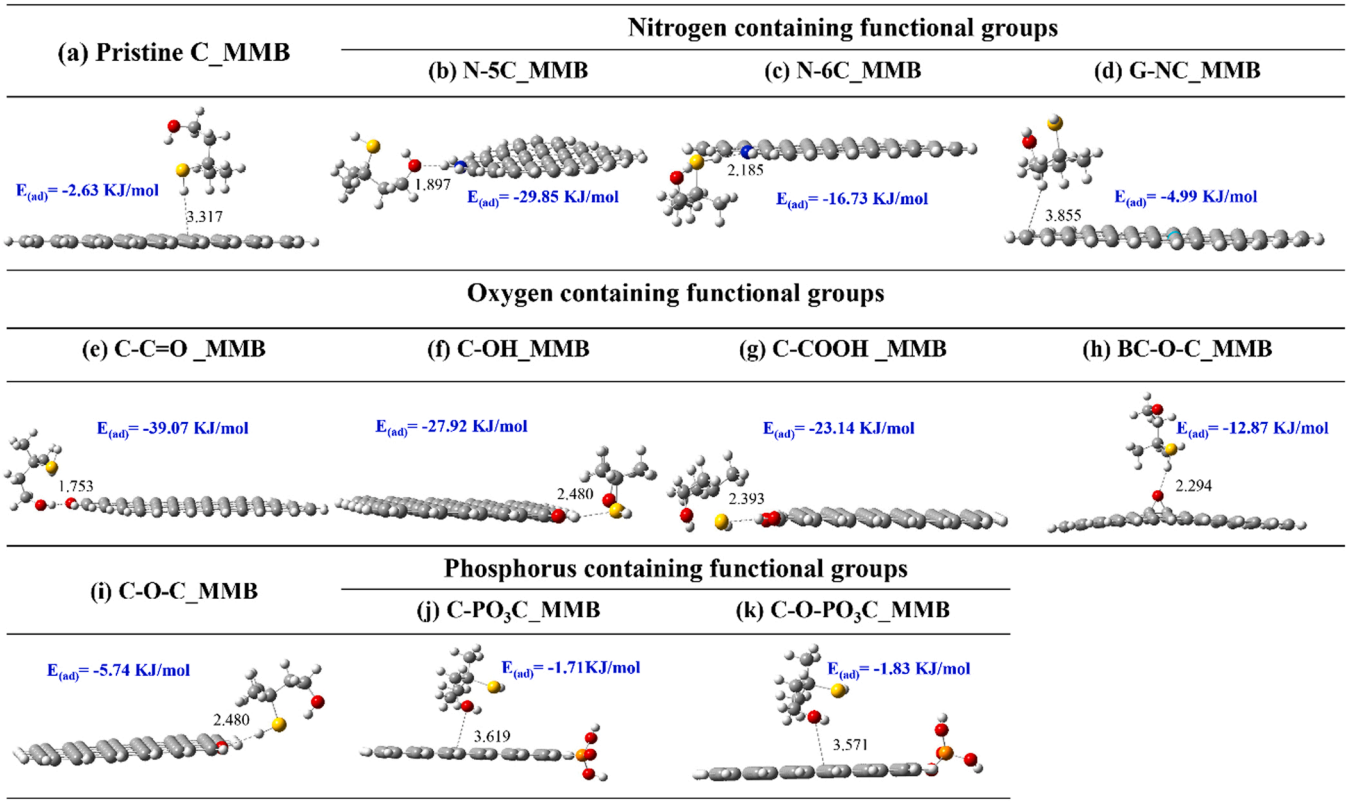



**Fig. 5.** The most stable MMB adsorption configuration in each surface model.

### 3.3.2. Adsorption configuration and adsorption energy

To comprehensively study the interactions between the MMB molecules and the AC-SNP carbon surface, eleven configurations of pristine C_MMB, -COOH_MMB, C═O_MMB, C-OH_MMB, bridging C-O-C_MMB (BC-O-C_MMB), in plane C-O-C_MMB, N-6C_MMB, N-5C_MMB and G-NC_MMB, C-$PO_3$_MMB, C-O-$PO_3$_MMB are considered. The optimized configurations are listed in Fig. 5(a-k), and the binding energy or interaction energy were calculated and included in each of the configurations.

In the pristine graphene model, the electrophilic region of MMB were perpendicular to the carbon surface due to the electrostatic interactions (Fig. 5a), the adsorption energy for the configuration is − 2.63 kJ/mol. For the C-$PO_3$_MMB and C-O-$PO_3$_MMB models, the nucleophilicity region of MMB were perpendicularly positioned to the neutralized carbon surface (Fig. 5j and k) and show decreased adsorption energy (−1.71 and −1.83 kJ/mol for C-$PO_3$ and C-O-$PO_3$) compared with the pristine graphene model.

Considering the N atom functionalized graphene models, the highest adsorption energy was identified with the N-5C_MMB configuration (−29.85 kJ/mol), where the electronegative region of MMB was close to the pyrrole N to form relatively strong adsorption, and H-bonding should also be considered (Fig. 5b). N-6C_MMB configuration shows a relatively lower adsorption energy of − 16.73 kJ/mol, where the main force is from the electrostatic interactions between the weak positivity around the -SH group and the negatively charged pyridine N (Fig. 5c). While for the G-NC_MMB configuration, the MMB molecule was perpendicular to the carbon surface with the positively charged region floating over the negatively charged benzene ring region (Fig. 5d), and having a much smaller adsorption energy (−4.99 kJ/mol).

For the graphene models with the O-containing functional groups, MMB molecules are close to the functional groups from the side for most of the configurations. Specifically, the most stable structure for C -COOH_MMB is shown in Fig. 5(g), where the covalent bond between the S and C atoms in MMB is cleaved, and the free sulfhydryl spice is close to the electronegative C═O in -COOH. The adsorption energy for this configuration is − 23.14 kJ/mol whereas the highest adsorption energy of − 39.07 kJ/mol was obtained with C═O_MMB. The main force could be the static interaction between the strong and weak positive regions (-OH and -SH in MMB) and the electronegative region (C═O) in the graphene phase. H-bonding between the -OH and the C═O is also required to form the relatively strong interactions (Fig. 5e). The adsorption energy of the C-OH_MMB configuration is − 27.92 kJ/mol, which could be due to the similar interactions as that in C═O_MMB configuration (Fig. 5f).

Two types of C-O-C modified graphene models were studied, in the BC-O-C_MMB configuration, MMB is positioning perpendicular towards the bridging oxygen atom (Fig. 5h), the adsorption energy was − 12.87 kJ/mol, while interactions between MMB and in plane C-O-C could be H- bonding between the -SH and O atom (Fig. 5i), where a relatively small adsorption energy was obtained (−5.74 kJ/mol).

O and N containing functional groups especially C═O, -OH, -COOH, and N-5, N-6 modified graphene models have higher adsorption energy than the pristine modified graphene models, which indicated that the adsorption performance may have been improved by the oxidation/nitration process. AC-SNP has a slightly higher absolute content of C═O, -COOH, N-5, G-N and slightly lower C-O-C/C-OH and N-6 content than AC-NP, but the adsorption performance was far better than AC-NP, which is due to the higher specific surface area and stronger bonding between the functional groups and the MMB molecules. Results from this study have indicated that it is crucial to obtain to carbon materials with high specific surface areas, and high concentrations of functional groups particularly C═O, N-5, -OH, -COOH and N-6 to facilitate the MMB removal and maintain its stability in carbon materials.

The interactions between the functionalized graphene models and MMB are classified as physical adsorption if the adsorption energy is < −30 kJ/mol and belongs to chemisorption behavior if the adsorption energy is higher than − 50 kJ/mol (Wu et al., 2021). However, for both the pristine graphene model and various functionalized graphene models, the adsorption energy are all negative and less than − 50 KJ/mol, indicating more of a physical adsorption rather chemisorption as shown by the adsorption models. This contradiction may not be so because of the likelihood of synergetic effects among the multiple functional groups which will result in higher adsorption energy for chemical adsorption process, as the DFT calculations were based on individual functional groups and the MMB molecule.

### 3.4. AC-SNP in vitro cytotoxicity and the odor removal bio-cat litter formulation

In vitro cytotoxicity tests were performed to evaluate the safety of AC-SNP to be used in cat litter formulations. The GES-1 cell viability after incubation with different concentrations of AC-SNP extracts for 72 h are shown in Fig. 6(a). The cell viability was all over 95%, with slightly decrease from 99.5% to 96.5% when the concentration of AC-SNP extract increased from 10 to 200 mg/g. Compounds that show a cell viability of over 70% are considered as nontoxic (Badano et al., 2019). Therefore, using AC-SNP as the odor removal component in cat litter formulations is biocompatible and nontoxic, and was used as the odor removal ingredient for the sugarcane bagasse based cat litter formulations. Optimized odor removal bio cat litter formulation is made up of 7.5% guar gum, 8% modified lignin, and 2% AC-SNP (Fig. S7). The detailed optimization process is provided in Supplementary 2.2. The

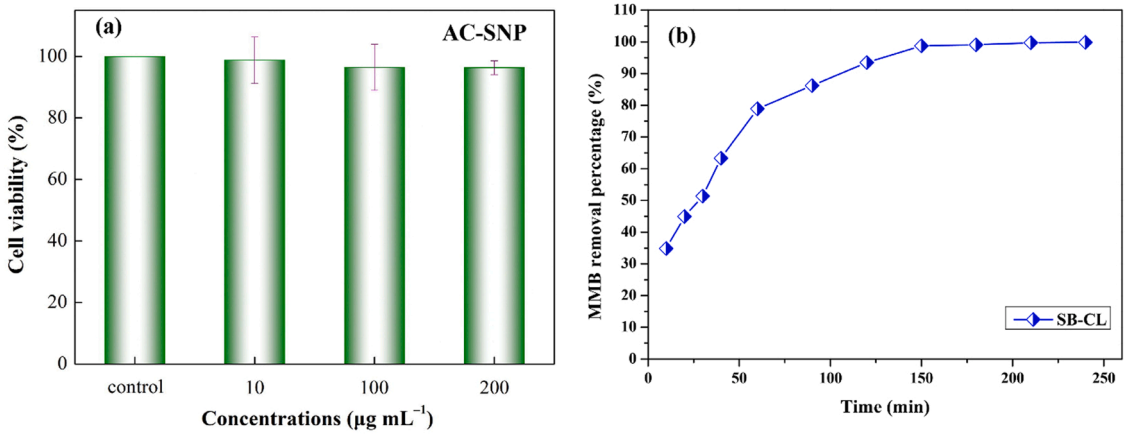



**Fig. 6.** (a) GES-1 cell viability after co-cultivation with AC-SNP for 72 h at different concentrations, and (b) Effect of the contact time on the adsorption of MMB onto the optimized bagasse-based odor removal cat litter (SB-CL) at initial concentration of 1.09 mg/mL.

**Table 6**
Properties of the sugarcane bagasse-based bio-cat litter (SB_CL).

| Cat litter | Odor removal (%) | *Salmonella* inhibition (%) | *Escherichia coli* inhibition (%) | Clumping percentage (%) | Hydration capacity (g $H_2O$/ g dry matter) |
|---|---|---|---|---|---|
| **SB_CL** | 99.87 | 97.58 | 98.31 | 86.83 | 6.20 |
| **Tofu cat litter** | 51.12 | None | None | 85.27 | 2.92 |
| **Bentonite cat litter** | 58.12 | None | None | 88.24 | 2.09 |

adsorption performance of SB-CL for MMB was evaluated from 10 to 240 mins, a slightly different time as the carbon materials. The amount of MMB removed by the SB-CL formulation was 34.86% by 10 min which is half the amount removed by the carbon AC-SNP on its own at the same time, though approached equilibrium at 180 min (Fig. 6b). The properties of the SB-CL were compared with tofu cat litter and bentonite clay litter, results are provided in Table 6, except for the comparable clumping percentage (86.83), the hydration capacity (6.20), odor removal efficiency (99.87%) and bacteriostatic rate (CFU log reduction of 97.58% and 98.31% towards *E. coli* and *Salmonella,* respectively) of SB_CL are much higher than both tofu cat litter and bentonite clay litter. This means that the cat litter formulation developed in this study could be advanced to the next of evaluation for commercialization.

## 4. Conclusion

A single-step preparation of a carbon material with increased porosity and specific surface area as well as increased O- and N-containing functional groups was developed by impregnating sugarcane bagasse with mixed $H_3PO_4$/$HNO_3$ under ultrasonic irradiation prior to carbonization. Ultrasonic irradiation was found to increase and stabilize the heteroatoms N and O in the treated biomass sample, and when the sample was carbonized at an elevated temperature the formed carbon inherited enriched N and O sites (O and N content of 13.48% and 4.04%, respectively). The physical and sonochemical effects of ultrasonic irradiation improved the etching of the biomass cell walls and the carbon material produced showed increased porosity (1.3541 $cm^3 \cdot g^{-1}$) and specific surface area (1603.1 $m^2 \cdot g^{-1}$), which facilitated the MMB sorption into the carbon matrix. The adsorption isotherm was better described by the Freundlich model. Kinetic studies showed that the adsorption of MMB onto AC-SNP follows the PSO kinetic model and the process involved was chemisorption. DFT calculations revealed that the adsorption energy between the functional groups and MMB molecules are in the order of C═O > N-5 > -OH > -COOH > N-6. The cat litter formulation containing extremely low ratio of 2% AC-SNP was found to effectively remove 99.87% of the MMB, and performed better than the commercially available cat litter formulations of tofu and bentonite clay. The adsorbent, AC-SNP was found be safe and nontoxic, and the cat litter formulation developed with it shows effective anti-bacterial properties against *E. coli* and *Salmonella* with CFU log reduction of 97.58% and 98.31%, respectively. As a result, the fabricated cat litter formulation has shown a significant commercialization prospect and should be taken to TRL (technology readness level) 4–5 stages.

## CRediT authorship contribution statement

**Shuo Wang:** Methodology, Investigation, Data curation, Formal analysis, Writing – original draft. **Caifeng Xie:** Supervision, Conceptualization, Visualization, Formal analysis, Writing – review & editing. **Fangxue Hang:** Supervision, Software, Methodology, Conceptualization. **Shupei Wang:** Supervision, Formal analysis, Validation. **Wen Li**: Methodology, Writing – review & editing. **Kai Li:** Conceptualization, Validation, Funding acquisition. **Anthony Mann:** Formal analysis, Software. **Sarina Sarina:** Methodology, Writing – review & editing. **William Doherty:** Methodology, Writing – review & editing. **Changrong Shi:** Supervision, Methodology, Formal analysis, Writing – review & editing.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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## Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.indcrop.2022.115493.

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